

Use the tabs on the left to access different sections of this document. This is ordinarily a living document; however a snapshot of it has been copied for showcase on Gage Magar's portfolio. Some sections have been omitted for the sake of secrecy.

Elevator Pitch

Experience survival-horror within the frozen, contorted remains of a post-apocalyptic city. You play as a survivor on a journey to reach the Fountain in hopes of defying death. Scavenge for supplies, fight immortal abominations, and stay warm to survive the eternal blizzard.

Design Pillars

1. **Survival:** It should feel like a desperate struggle through a nightmare. However, the player should not be defenseless. The player should feel like they are actively surviving and defending themselves.
2. **Dread:** The game should instill a sense of slow, uncanny, surreal horror. An overarching feeling of dread should be prioritized over jumpscares.
3. **Decision-making:** Almost every resource or choice in the game should be weighed against a multitude of factors. This helps add a great deal of interest and strategizing to gameplay, along with keeping the player constantly second-guessing themselves.

Gameplay Mechanics

Core

- **Resource Management:** The player's inventory is a grid, and items take up varying amounts of grid spaces. The player is constantly collecting and using items, and needs to determine when and how to effectively pace their usage. Using items now will make things easier now, but harder later. Saving items will make things harder now, but much easier later
    - If the player doesn't have enough room for items, they'll have to either rearrange their inventory (kind of like a mini game), or they'll have to drop something
    - **Temperature:** Temperature is a constantly draining resource while outside. If it reaches 0, the player dies. They can regain temperature by lighting fires using

matches. Fires are permanent and quickly restore temperature. While indoors, temperature drain stops, but doesn't warm up. This means the player is still encouraged to search for sources of heat inside buildings when they first enter
        - **Matches:** One of the most important, core resources. Used to light fires, as well as permanently kill downed enemies
    - **Health:** The player needs to use items to heal, making health a more precious resource
    - **Stamina:** A short-term resource used in combat
    - **Crafting:** Some items can be combined to make new ones. Crafting is also encouraged since it frees up inventory space. However, the player might choose to hold on to crafting resources so as to use it to make something better
    - **Weapon Durability:** Melee weapons have durability (meanwhile ranged ones require ammo). Durability is the main reason for the player to choose stealth over combat in certain situations
    - **Save Points:** Creating a permanent save point requires an item, which adds to the decision-making process

Secondary

- **Combat:** The main tool for dealing with enemy encounters. Combat consumes resources, such as weapon durability, health, and/or ammo. Combat is intended to feel satisfying to execute well, but shouldn't make the player feel empowered, as this would reduce the feeling of horror. Positioning and space can heavily influence the feel of an encounter
    - **Burning Enemies:** Most enemies don't stay dead when knocked out, instead reviving with reduced health after some time. The player must use matches to burn the body after a combat encounter
    - **Stamina:** Running, dodging, and attacking all consume stamina. Stamina influences the pace of combat, as they can't just stand there and attack indefinitely. Eventually, the player will be forced to fall back to regain it
    - Enemies don't have knockback on normal attacks. This is to ensure they stay intimidating, as the player is always the one being pushed back instead of the enemies
    - **Dodging:** The player has the ability to dodge, which quickly moves them out of the way of an attack. This is helpful in tough situations where damage would otherwise be unavoidable
    - **Hands:** Usually, the player would have a light source in their offhand and a weapon in the main hand. However, stronger weapons require two hands to wield, which means the player must decide whether to prioritize sight or damage

Supporting

- **Navigation:** The world layout is complicated and interconnected, but should encourage exploration. Maps are the main tool the player uses to navigate. Maps don't immediately tell the player's current whereabouts, though they do automatically update with markers

when the player discovers locked doors and important locations. The map also provides a reminder of the current objective
    - **Signposts:** When interacted with, signposts indicate the player's position on the map
    - **Compass:** Indicates the player's facing direction
- **Stealth:** The secondary tool for dealing with enemy encounters. The player may choose to use stealth as it consumes less resources than combat. However, it may not be viable in every situation
    - **Sightlines:** If the player enters the enemy's sightline, they get alerted and chase them
    - **Sneaking:** The player can sneak. While sneaking, enemies can't detect the player from behind, even if they're right up against them
    - **Sounds:** Certain loud actions the player takes may lure enemies (IE: firing a gun). The player may choose to throw an item to lure an enemy out of the path of progression

Gameplay Loop

- **Main Areas:** There is a constant back-and-forth sense of pacing created by exterior and interior areas. Interiors are more tense, while exteriors add a bit of leeway
    - **Exterior Areas:** Tend to be more open/large and shift focus to temperature management
        - Find a signpost to determine whereabouts
            - Use the map to determine a route and plan of action
        - Manage temperature and find a balance between making progress and not freezing
            - Find new places to light fires or return to old ones
            - May decide to return to interior areas to search for warmth
            - Lighting fires slowly makes the world easier to navigate, especially when the player later returns to previous areas (creates a sense of "conquering the level")
        - Avoid, fight, or stealth around enemies that emerge from the fog
            - Requires resource management
        - Search for the next objective or for rewards in hidden side paths
    - **Interior Areas:** Tend to be shorter
        - Cautiously maneuver through level
        - Scavenge searchable objects for supplies
        - Avoid, fight, or stealth around enemies
            - Requires resource management
            - Combat is much more dangerous due to interiors usually being tight and claustrophobic
            - More handleable situations should encourage combat, more dangerous situations should encourage stealth
            - Killing enemies adds to the sense of "conquering the level"

- **Eidolon Encounter:** A recurring miniboss designed to test the player's skills with navigation, temperature management, and stealth. This boss is made easier if the player spent more time "conquering" the level earlier
    - Find a place to hide, waiting for the Eidolon to pass
    - Quickly find a signpost to determine whereabouts
        - Use the map to quickly determine a route and plan of action
    - Manage temperature and find a balance between making progress and not freezing
        - Places of warmth are often placed away from hiding spots to create tension
- **Dungeon Areas:** Larger interior areas that test the player on how many resources they've saved up through the main area of the game
    - Matches the gameplay loop of interior areas, but on a larger scale
    - **Boss Encounter:** Each encounter is designed to test the player's combat and resource management skills, as well as provide a climactic conclusion

Additional Content

- Levels should have additional side areas to explore for extra lore or items
- Achievements that encourage unique playstyles
- Simple visual easter eggs in levels

Art Style

- The game is 3D with a retro PS2 art style, featuring pixelated textures and low-poly models
- There are some modern rendering features though, such as proper lighting, shadow mapping, post process effects, PBR materials, volumetric fog, etc.
- This is done because low-resolution models/textures force the player's mind to fill in more details (the human mind is usually better at scaring itself). Additionally, this will make the development process much quicker

Music Style

- Overall, the soundtrack should feel uncomfortable (because of the nightmarish horrors of the blizzard) and ethereal (because of the ghostly, somber nature of the story). It should also feel reminiscent of the cold, snowy, icy environment the game is set in
- Most tracks will be slow, atmospheric, unnatural, and distressing. These atmospheric tracks should feel unpredictable and unique to set the player on edge. However, not all areas or scenarios will have music, since sometimes silence can be more effective
- Some tracks (such as boss fights) will be more distinct, with painful instruments and cacophonous symphonies to stress the player
- Some rare tracks will be calming and melodic to provide respite from the horrors. They should feel relieving and mix in elements of the ethereal atmosphere

- Some scenarios will have adaptive music, where the main area theme will be replaced. For instance, if the player is about to freeze, the music should be tense to reflect that

Marketing

Audience

- **Age Range:** 18 - 50
- Target demographic is those more experienced with challenging video games
- Target audience is people interested in retro-style horror and/or survival-horror games
- **Older audience range (30 - 50):**
    - Target **nostalgia** as a selling point
    - Nostalgia is featured in the art style, as well as in references to old survival-horror games (IE: Silent Hill save screen and Resident Evil doors)
- **Younger audience range (18 - 30):**
    - The game makes for great reaction content for streamers and creators. This is where we could hit the younger range

Similar Games

These can be looked to in terms of audience or marketing strategy

- Resident Evil series
    - Resident Evil 4
    - Resident Evil 7: Biohazard
- Silent Hill series
    - Silent Hill 2
    - Silent Hill 3
- The Forest
- Cry of Fear
- Lost in Vivo
- Amnesia: The Bunker
- The Last of Us
- The Long Dark

Release Structure

- Based on some general research, indie horror games tend to be priced somewhere around the $10 to $20 range
- **Base Game**
    - Price will be $8 ($7.99)

Marketing Phases

Marketing is split into a series of phases.

1. **Phase 1 (Introductions):** Start introducing the game to a wider audience using posts that describe what the game is (IE: what is the gameplay? The story? The visuals? The history behind it?)
    - Aim for 5 major posts for this phase
    - **Steam Page:** The Steam page will be released some time during phase 1
    - **Gameplay Trailer:** The opening of the Steam page will coincide with the trailer dropping:
        - This will be more like a teaser trailer
        - The game might not be totally complete yet, hence this trailer will not be the main one (which will be created later)
2. **Phase 2 (Demo Release):** Once an audience is generated, post more varied content to keep people engaged. This could, for example, show off parts of development and be a little less formal. It should also begin to hype up the demo release
    - **Main Trailer:** This will be created once development of the base game is nearing completion
        - Should hype up the base game's release and mention the demo if applicable
        - Have a focus on both gameplay and story
        - This should include a **hard release date deadline**
    - **Demo:** This phase will end with the release of a short, free demo (only for base game). The demo will include the entire opening section, concluding with the town intro cutscene (around 20 minutes of gameplay)
3. **Phase 3 (Base Game Release):** Push the marketing further with the goal of gaining wishlists and eventual conversion on sales. This phase begins with the release of the trailer and ends with the game released
4. **Phase 4 (Content Creator Push):** After the base game is released, we need to continue pushing the game with the goal of conversion and retaining engagement. The primary method of doing this will be through pushing the game to streamers and content creators
    - **Note that this phase might partially overlap with the previous one. Codes to the game should be given to content creators before release**
        - This also gives extra incentive to creators
    - Start by pushing the game to **small streamers on Twitch and YouTube**
        - If we get it popular with smaller accounts, that gives it the potential to spread upwards to larger ones
    - Give out free codes
    - Contact smaller channels
    - Ask any friends with channels to try the game
    - If funds permit, pay some streamers to play it
    - During this time, the game may continue to receive updates

Social Media

- The amount of posting that will need to be done varies per platform

- The general structure is that there will be weekly “major” posts, which occur as a wave across all platforms. While they will have the same premise and use the same assets, the editing cannot be the same across all. There will be two primary formats for video posts:
    - **Primary form:**
        - Between 0:45 to 2:00 long
        - Used on Reddit, X, and Bluesky
        - Having a strong hook in the first 5 seconds is important
    - **Short form:**
        - Between 0:30 to 1:00 long
        - Used on Instagram, YouTube shorts, and TikTok
        - Having a strong hook in the first 5 seconds is **even more** important
        - Captions need to be **intriguing without being “clickbaity”**
            - Prompt the viewers to respond
            - Be more personable
- If time does not permit for creating a video post for a week, image posts may be done instead (can be done as slideshows in TikTok)
- **Content of posts:**
    - **Always post a link to the Steam page! Wishlists and conversion are the goal! Make the link as readily accessible as possible!**
    - Find a memorable thing about the game to market. In our case, it could be “a horror game where fire is vital to survive the cold and kill the undead inhabitants of an eternal blizzard”
    - Being intentionally subtle or mysterious about the nature of the game is often a bad idea
    - Remember to pace the content of the posts. Don’t show off too much of the game, but don’t be too repetitive either

Reddit

- Video and image posts
- Posting schedule on Reddit should be less frequent to avoid spamming (1 post per week on average)
    - Post around 12:00 PM to 1:00 PM on weekdays
- We’ll want to post the same post across different relevant subreddits to maximize reach
- **Primary focus subreddits:**
    - r/survivalhorror
    - r/IndieHorrorGaming
    - r/ps2_graphics
    - r/low_poly
    - r/IndieGaming
- **Secondary focus subreddits:**
    - r/IndieGameWishlist
    - r/SurvivalGaming (only if post is relevant to survival mechanics)
    - r/Unity3D (only if post is relevant to technical implementation)

    - r/gameDevMarketing
- Types of posts that tend to be more successful:
    - Ones that directly encourage engagement (ask people what they think)
    - Posts that follow a relevant trend
    - Posts that are written to feel authentic and personable (IE: don't be afraid to write like you're a developer posting about the game, as opposed to a faceless company)
- Create a delineation between two types of posts:
    - Formal reveals (trailer drops, important updates, etc.)
    - Informal updates (updates on how the game is going that invite feedback)
- **Reddit can be a great place to gauge interest and receive feedback directly from your target audience**
- **Respond to comments!**
- Don't be afraid to make it clear that this is being developed by a team. Based on a researched game, people are more likely to take these projects seriously

Instagram

- Video and image posts
- **Focus on reels**
- Post on weekdays during the afternoon
- Success requires interacting more with other relevant accounts, leaving likes, posting comments, etc. This requires some time investment
- Types of posts that tend to be more successful:
    - Simplify the game. Keep it brief

X

- Video and image posts (but videos/gifs are more successful)
- Posting schedule should be more frequent (at least once every other day, but daily is even better)
    - When time doesn't permit, filler text posts can be made
    - Reposts of existing content is an even better way to retain engagement
    - **Post at peak times: early morning and late evening**
    - 3 posts per week absolute minimum
- Types of posts that tend to be more successful:
    - Hook people with questions
    - Less serious posts can work better here
- **Use tags:**
    - #survivalhorror
    - #indiegame
    - #horrorgame
    - #gamedev
    - #retro
- **Respond to comments!**

- The pinned post should be our most engaging one
- **Participate in informal events**
    - #screenshotsaturday
    - #indiedevhour
- Steam key giveaways (once the game is released)
    - Ask for retweets and comments as entry
- Could potentially reach out to advertising accounts (IE: https://x.com/clemmygames, IndieGameJoe, IGN Game Trailers)

YouTube Shorts

- Video posts
- Main channel feed only used for important videos such as teasers and trailers
- Optimal posting times are Tuesday through Friday, between 3:00 PM and 7:00 PM
- **Respond to comments!**

Bluesky

- This is an inherently smaller platform, so less engagement is expected
- Video and image posts (but videos/gifs are more successful)
- Posting schedule should be more frequent (at least once every other day, but daily is even better)
    - **Post at peak times: early morning and late evening**
- **Use tags:**
    - #survivalhorror
    - #indiegame
    - #horrorgame
    - #gamedev
    - #retro
- **Utilize feeds:**
    - Horror Games Community
    - Indie Horror Games
    - Indie Devs & Games
- **Respond to comments!**
- The pinned post should be our most engaging one

TikTok

- Video posts
- Posting schedule should be more frequent (at least once every other day, but daily is even better)
    - Optimal to post between Tuesday and Thursday between 2:00 PM and 6:00 PM
- Types of posts that tend to be more successful:
    - Encourage people to watch to the end. The algorithm rates videos higher when people watch to the loop point

    - Less serious posts can work better here
    - It doesn't have to be over-edited as long as the hook is catching and content keeps people watching
- **Respond to comments!**
- **Use tags:**
    - #survivalhorror
    - #horrorgame
    - #gamedev
    - #retrohorror
    - #indie
- Include important keywords in the text descriptions
- **Playtester and content creator reactions are a great source of content!**
    - Ask playtesters if they are fine with being recorded
    - Would have to consult the content creators directly

Levels

Outskirts (LevelOutskirts)

- A mountainous, forested area along the edge of town. The paths wind down in a spiraling manner, reflecting how the player is delving deep into dangerous territory
- Tutorializes basic movement, interacting, temperature, notes, locked gates, maps, signposts, doors, and saving
- Introduces Lantern, Bundle, Notebook, Map, Matches, Heavy Board, Sanguine Powder, and Crafting Scrap items
- Features a set piece of frozen deer
- **Old House (LevelOldHouse)**
    - Tutorializes weapons, scavenging, healing items, hotkey swapping, combo locks, and combat
    - Introduces Ice Pick and Medkit items
    - Introduces Husk enemy
- **Welcome Center (LevelWelcomeCenter)**
    - Tutorializes sneaking, staggering enemies, burning downed enemies to kill them, and crafting
    - Introduces Canned Food item
    - Includes a short stealth section to foreshadow the fact that seemingly frozen Husks can come to life

South Aurburg (LevelTown)

- A smaller, low-density region in the city of Aurburg. It should look like an old American boom town mixed with modern infrastructure. Many regions are frozen over and snowed-in. The Claret River separates this side of town from North Aurburg
- Introduces Long Pipe, Axe, Bottle of Alcohol, and Molotov Cocktail items
- Introduces Sewer Arm enemy
- Various parts of town are broken and collapsing into a deep abyss. This abyss has lore implications (it was created when the Fountain erupted), as well as blocks the player from leaving the area
- **Corner Stop (LevelConvenienceStore)**
    - Tutorializes stealth, thrown items, weapon durability, empty can traps, and boarding up doors
    - Introduces Empty Can item
    - Introduces the first padlock, but the player doesn't have a pistol to shoot it yet
- **Dante's Italian Cuisine (LevelRestaurant)**
    - Tutorializes glass traps and the ability for Husks to play dead
    - Features a secret space accessible via a padlock
- **Aurburg Daily**

    - Optional side area to unlock an extra path that's handy during the Eidolon sequence
    - Includes lore as a reward
- **South Aurburg Police Department (LevelPoliceStation)**
    - Tutorializes ranged combat and flares
    - Introduces Hand Flare and Pistol items
    - Introduces Face Arm Husk and Corpse Bird enemies
    - Is split into three distinct sections:
        - While the generator is on, the building it safe to explore
        - Turning the generator off turns everything dark and dangerous, awaking enemies
        - Obtaining the pistol makes the area feel safer again
- **Local Peak Marketing**
    - Tutorializes gas cans
    - Playground for the Pistol weapon just obtained
    - Optional side area
- **Lionhart West Apartments (LevelApartments)**
    - Introduces Pill Bottle item
    - Features combinations of stealth and combat with several Husks
    - Includes a set piece with a creepy looping apartment to add lore
- **Aurburg Historical Society (LevelMuseum)**
    - Introduces Eidolon enemy
    - Features a set piece where the player must escape through a hallway packed with Husks

Lionhart Hospital (LevelHospital)

- A "dungeon" area
- The blizzard's influence has warped the hospital, creating irrational architecture (IE: upside-down rooms, rooms with pits leading into darkness, endless hallways, etc.)
- Includes unique combinations of stealth and combat scenarios to test the player
- **Morgue (LevelMorgue)**
    - Appears as a massive tower of morgue cooler doors with an infinite ceiling
    - The floor is connected to the boss. It is a giant mass of melded bodies
    - The boss encounter involves combat with several enemies

Enemies

Common

- **Husk**
    - **Appearance:** Extra tall humanoids with bluish-gray, frozen skin and a bottomless hole in place of a face. They twitch and writhe unnaturally while moving
    - **Mechanics:**
        - Runs after the player if they go too far out of range or shoot with a gun before they detect the player. Meant to feel unpredictable
        - Can play dead, blending in with static/frozen/inanimate corpses. Awakens when the player triggers a trap or manually placed trigger
        - Can dodge
        - Can stagger
        - Can open doors
    - **Attacks:**
        - **Standard swipe:**
            - Low damage
    - Medium health
    - Slow walk speed
    - Fast run speed
    - Medium aggression
- **Husk (Aggressive Variant)**
    - **Appearance:** Extra tall humanoids with bluish-gray, frozen skin and a bottomless hole in place of a face. The hole is filled with a strange mass dotted with trypophobic holes. They twitch and writhe unnaturally while moving
    - **Mechanics:**
        - Runs after the player if they go too far out of range or shoot with a gun before they detect the player. Meant to feel unpredictable
        - Can play dead, blending in with static/frozen/inanimate corpses. Awakens when the player triggers a trap or manually placed trigger
        - Can dodge
        - Can stagger
        - Can open doors
    - **Attacks:**
        - **Standard swipe:**
            - Low damage
    - Medium health
    - Fast walk speed
    - Fast run speed
    - High aggression
- **Husk (Face Arm Variant)**

    - **Appearance:** Extra tall humanoids with bluish-gray, frozen skin and a bottomless hole in place of a face. Before the arm emerges, they look like normal husks, making them unpredictable. A long arm emerges from their face and is their method of attack. They move as if being controlled or puppeted by the arm
    - **Mechanics:**
        - Runs after the player if they go too far out of range or shoot with a gun, but not while the face arm is out
        - Face arm emerges when the player is near
        - Can dodge
        - Can stagger
    - **Attacks:**
        - **Sweep:**
            - High damage
            - Does not rotate to face the player, so this is an opportunity for the player to run around and attack it in the back
        - **Down swipe:**
            - High damage
            - Rotates to face the player, making it almost guaranteed to deal damage if the player is too close
    - Medium health
    - Slow walk speed
    - Fast run speed
    - Low aggression

Uncommon

- **Sewer Thing**
    - **Appearance:** Is a long arm that emerges out of sewer drains. It has tumorous fingers branching off of other fingers, making it look akin to roots. It snaps and pops as its joints move
    - **Mechanics:**
        - Rises when the player approaches and hides after a while if nothing is detected
        - The player can waste ranged ammo to kill them and make street navigation easier, but there's always an alternate route to avoid them entirely
    - **Attacks:**
        - **Sweep:**
            - High damage
            - Cannot be dodged past
        - **Stab:**
            - High damage
            - Cannot be dodged past

            - Stays in the extended position long enough that the player can use melee to attack it
    - Low health
    - Stationary
    - Low aggression
- **Corpse Bird**
    - **Appearance:** Appear as pterodactyl/vulture-like birds made from fused, gray, human corpses. Their wings are made from fingers mimicking feathers
    - **Mechanics:**
        - Screech upon first noticing the player
        - Can stagger if shot in the mouth while screeching or vomiting. This causes them to collapse to the ground, giving the player an opportunity to use melee
    - **Attacks:**
        - **Screech:**
            - Drains player temperature
            - Alerts nearby enemies
        - **Vomit:**
            - Creates acid splashes on the ground in a forward line that remain for a few seconds
            - Moves forward during the attack, forcing the player to move out of the way and adjust their aim
            - Low damage
    - Medium health
    - Slow speed
    - Medium aggression

Stalking

- **Eidolon**
    - **Appearance:** Is a large, two-headed, four-armed shadow creature with a hole in its chest. Its surface is made up of swirling, tortured faces and several writhing arms emerge from its shadowy base. Its face has a hole from which a bright blue light emerges, indicating where it can see
    - **Mechanics:**
        - At set points, it will appear to hunt the player down
        - When it arrives, time and enemies will literally freeze in place
        - Portals are created at spots on the ground when it appears
        - It is always patrolling along a set path. Set nodes will cause it to play an inspecting animations, and others will cause it to use a portal to teleport somewhere
        - Has a large hearing range
        - Temperature drain greatly increases while it is present

        - Can see through the face of living enemies. When a player enters a frozen enemy's line of sight, it will alert the Eidolon
        - If the player enters its line of sight, an intense chase will commence
            - It can teleport through portals and re-emerge near the player if it determines it would be faster to do so
            - It can attack through portals placed on the walls
            - Causes visual distortions while present
    - **Attacks:**
        - **Hug:**
            - Instakill damage
        - **Wall Portal Swipes:**
            - Medium damage
    - Unkillable
    - Slow walk speed
    - Fast run speed
    - High aggression

Bosses

- **The Mortician**
    - **Appearance:** Appears as a large humanoid on the upper half, while the lower half is fleshy and melded with the floor of the morgue. The floor consists of a massive pile of bodies fused together, all of which are ultimately part of the same creature. It has hands that can burrow through and reach out through the floor.
    - **Mechanics:**
        - Swipes at the player if they get near
        - If the player is standing still for too long, an arm emerges to grab them, of which they must attack to get free
        - Between phases, pillars emerge, spewing cold air. The player has to rush to hit weak points to collapse the pillars and stop themself from freezing
        - As phases progress, it summons Sewer Things that attack in a pattern
    - **Attacks:**
        - **Standard Swipe:**
            - Medium damage
        - **Grab:**
            - Temporarily immobilizes the player
    - Medium health
    - Stationary
    - Low aggression

Items

Miscellaneous

- **Lantern (Lantern)**
  - A handheld kerosene lantern that the player keeps throughout the whole game
  - Cannot be dropped
  - Serves as a light source
  - Held in the left hand
- **Notebook (Notebook)**
  - Stores previously collected notes
  - Cannot be dropped
- **Map (Map)**
  - Needs to be collected as an item first before it can be used to navigate an area
  - Cannot be dropped
- **Bundle (Bundle)**
  - A story item the player has on them the entire game
  - Cannot be dropped
- **Matches (Matches)**
  - Can be used to ignite fires and burn downed enemies
  - Common
  - The most important resource in the game
- **Crafting Scrap (Scrap)**
  - Used solely for crafting
  - Common
  - Combine with Crafting Scrap to make Matches
  - Combine with Bottle of Alcohol to make Molotov
- **Empty Can (EmptyCan)**
  - Can be thrown to create a distraction
  - Common
  - Combine with Empty Can to make Scrap
- **Hand Flare (Flare)**
  - Used to provide light in an area, as well as temporarily slow temperature drain while held or stood near. Can be thrown
  - Uncommon
  - Useful when the player needs to unequip their lantern to two-hand a weapon
- **Sanguine Powder (SavePowder)**
  - Can be placed in stationary fires to create save points
  - Rare
  - Spawns are manually placed in levels
  - In the lore, this powder is known as “alkahest”

Melee Weapons

- **Ice Pick (IcePick)**
    - A slightly weaker weapon with low stamina cost
    - One-handed
    - Low damage
    - Medium speed
    - Medium stamina cost
- **Long Pipe (Pipe)**
    - A weak weapon with low stamina cost and long range
    - Two-handed
    - Low damage
    - Medium speed
    - Low stamina cost
    - Medium Inventory size
- **Heavy Board (WoodenBoard)**
    - A single-use weapon which staggers enemies upon hitting them. It can also be used to temporarily board up doors
    - Two-handed
    - Medium damage
    - Medium speed
    - Medium stamina cost
    - Large inventory size
- **Axe (Axe)**
    - A sluggish but heavy-hitting weapon
    - Two-handed
    - High damage
    - Slow speed
    - High stamina cost
    - Large inventory size

Ranged Weapons

- **Pistol (Pistol)**
    - A small pistol with generally low power
    - One-handed
    - Common ammo
    - Medium damage
    - Fast reload speed
    - Short range
    - Small inventory size
    - Ammo is named **Pistol Ammo (PistolAmmo)**
- **Bottle of Alcohol (Bottle)**

    - Can be thrown to create a puddle of flammable liquid. If shot, the liquid ignites. Flares will also ignite the liquid
    - Rare
    - Combine with Crafting Scrap to make Molotov Cocktail
    - Medium inventory size
    - Mainly just meant to be used to create Molotov Cocktails. However, it can be used on its own
    - Fire deals less damage than a Molotov Cocktail
- **Molotov Cocktail (Molotov)**
    - A throwable item which creates a fiery explosion, leaving a lingering area-of-effect. Requires matches to be used
    - Rare
    - Short range
    - Medium inventory size
    - Fire deals more damage than a Bottle of Alcohol on its own

Consumables

- **Canned Food (CannedFood)**
    - A less ideal healing item option. Is better when saved for cooking at a fire
    - Common
    - Low, fast-speed, gradual healing
    - Can be cooked using a fire
    - When consumed, adds an empty can to the player's inventory
- **Cooked Canned Food (CookedCannedFood)**
    - A generally good healing item with an added warmth boost
    - Medium, medium-speed, gradual healing
    - Provides temporary immunity to the cold
    - When consumed, adds an empty can to the player's inventory
- **Medkit (Medkit)**
    - A powerful healing item, but which lacks the warmth-boosting effects of cooked foods. Meant to be usable during combat due to its instant healing effect
    - Rare
    - High, instant healing
    - Large inventory size
- **Pill Bottle (PillBottle)**
    - A consumable which temporarily grants infinite stamina, but increases damage taken
    - Uncommon
    - Temporary infinite stamina
    - Temporarily multiplies any damage taken

Achievements

Progression/Required

- **Cold Embrace (ReachHospital):** Enter the hospital
    - Easy difficulty
    - Hidden
- **Post Mortem (MorgueBoss):** Defeat the morgue
    - Moderate difficulty
    - Hidden

Challenges

- **Blizzard's Miracle (MeatLocker):** Collect the item at the end of the meat locker
    - Easy difficulty
    - Hidden
- **Transitive Property (IgniteWoodenBoard):** Light an enemy on fire using a wooden board
    - Easy difficulty
- **Death From Above (SnipePoliceStation):** Down an enemy in the Police Department lobby from upstairs
    - Easy difficulty
- **Sharpshooter (ShootBottle):** Shoot a bottle in midair
    - Easy difficulty
- **Screams of the Damned (IgniteGasSplash):** Light three enemies on fire using the same Molotov Cocktail or gas puddle
    - Moderate difficulty
- **Quiet as a Shadow (ConvenienceStoreStealth):** Navigate the entire Corner Stop without getting noticed
    - Moderate difficulty
- **Skill Issue (EmptyCanKill):** Kill an enemy using nothing but an Empty Can
    - Moderate difficulty
- **Lend a Hand (SewerArmMelee):** Defeat the thing from the sewer using nothing but melee
    - Moderate difficulty
- **Imitations of Instinct (KillAllCorpseBirds):** Kill and burn every winged beast
    - Hard difficulty
- **Chronicler (AllNotes1):** Collect every note
    - Moderate difficulty
- **Gauntlet of Cold (NoMatches1):** Complete the game without using matches to light any fire pits
    - Hard difficulty

- **Untouchable (NoDamage1):** Complete the game without taking damage
    - Extreme difficulty
- **Completionist (Completionist):** Unlock every achievement
    - Extreme difficulty

Team Standards

Communication

- Communication occurs in a team Discord server and associated “dev” text/voice channels
- Team members should regularly provide progress updates for their work through either Trello or Discord. Posts should still be made to Trello for milestones on a task though, even if posting on Discord as well
- Voice or in-person meetings aren’t a consistently scheduled thing

Trello

- Trello should be kept up to date to reflect the current state of people working on tasks
- Tasks that are to be done in the future are kept in the “Story Backlog” list
- Tasks that should be done roughly within the next two weeks are kept in the “Current Sprint” list
- There are templates for commonly recurring tasks stored toward the left
- Tasks have color-coded headers based on the department or discipline. There is a color key in the Trello for this
- **When a person is assigned to a task, they are expected to manage it. The following should be done to keep tasks up-to-date:**
    a. When starting a task, move it to the “Doing” list
    b. Read the description for more details
    c. Make sure to check of checklist items as they are done
    d. Post updates on the task, such as comments and screenshots. This should especially be done when it is completed for approval
    e. Move it to the “Needs Approval” list once finished
    f. It may be sent back-and-forth if changes are suggested
    g. Check off the task itself once approved and move it to “Completed”

GitHub Repo

There are a strict set of rules to avoid merge conflicts

1. Always make sure to PULL each time you go work on something to ensure the repo is up-to-date
2. Avoid touching areas where you know someone else is working in
3. BEFORE pushing, always make sure to scan over your list of changes. If there’s something you didn’t intend to change, it might be best to revert it
4. Since Unity stores whole scenes in one file, multiple people working in the same scene has the potential to create massive merge issues. To avoid this, **use the “Scenes” and “Checked Out Scenes” lists in Trello:**

a. First check the Scenes list. If the scene is already in the Checked Out Scenes list, someone is currently working on it and it shouldn't be touched
b. If not checked out, you can move it over and add your profile to the card
c. When finished and pushed, make sure to move the scene back and remove your profile

Unity Engine Version

The version of Unity the game currently runs on is **2022.3.62f3**

File Organization

File Naming Conventions

Bar some exceptions, the general rule of thumb is that **pascal case should ALWAYS be used**, and one should **avoid including the type of file in the name** (IE: appending "Texture" to a .png file is redundant since that's implied by the file type)

- **Folders:** Files are organized based on what they're used for, not what type of file they are
    - **Categorization Folders:** "CategoryName"
    - **Shared Folders:** _Shared
        - Used to group assets together that are shared by things in a category
        - Don't call something like _PropsShared, since that would be redundant. It would just be called _Shared. The fact that it's stored under the Props folder implies that it's shared files used by the props
    - **Asset Clump Folders:** "AssetName"
        - Folders that contain all the associated files for a particular asset
    - **File Type Folders:** _"FileType"
        - These are optional. Sometimes files are just dumped right inside asset clump folders
        - Important: these shouldn't contain ANY other folders inside of them!
        - _Textures
        - _Materials
        - _Shaders
        - _Meshes
        - _Animations
        - _Audio
        - _Prefabs
        - _Scripts
- **Files:** Usually named based on the asset clump they are associated with. **Avoid including the type of file in the name** (IE: appending "Texture" to a .png file is redundant since that's implied by the file type)
    - **Textures**
        - **Albedo:** "AssetName".png

        - **Normal:** “AssetName”_n.png
        - **Metallic/AO/Detail/Smoothness Mask:** “AssetName”_m.png
        - **Emission:** “AssetName”_e.png
        - **Special:** “AssetName”“SpecialMap”.png
    - **Materials:** “AssetName”.mat
        - **Object Space Surface Variant:** “AssetName”_ObjectSpace.mat
        - **Scale Surface Variant:** “AssetName”_Scale.mat
        - **Snowy:** “AssetName”_Snowy.mat
        - **Special:** “AssetName”“VariantName”.mat
    - **Meshes:** “AssetName”.fbx
    - **Prefabs:** “AssetName”.prefab

Folder Structure

Any root asset folders not listed here are associated with specific packages/plugins and probably shouldn’t be touched

- **GameBuilderCore:** Library containing re-usable code intended to be re-usable in other projects
- **GameBuilderCoreHDRP:** Same as above, but for HDRP-specific projects
- **Gizmos:** Contains editor-only gizmo icon graphics dynamically obtained from scripts
- **Marketing:** Contains scenes and assets never used in the game, just created to stage screenshots for marketing assets
- **Project:** The main folder for all game assets. Should usually only need to touch things in here
    - **_Shared:** A miscellaneous folder for things used throughout the game that don’t easily fit another category. Contains a few things like HDRP rendering settings
    - **Dialogue:** Contains audio and scriptable object configurations for dialogue
    - **Entities:** Contains code, prefabs, etc. for the player controller and all AI entities
    - **FunctionalProps:** Contains code, prefabs, and associated meshes/materials/textures for props that have functionality. Sometimes props have both a functional and non-functional version, in which case the meshes/materials/textures and static prefab are stored in the Props folder
    - **Items:** Contains code, prefabs, etc. for all items in the game
    - **Levels:** Contains the scene files for levels that the player can play in. Sometimes contains some asset files if the asset is dedicated to/restricted to that level and not re-used elsewhere
    - **Localization:** Contains all localization string tables
    - **PostProcessEffects:** Contains the shaders and textures used for custom post processing effects (IE: dithering, freezing overlay, damage overlay, etc.)
    - **Projectiles:** Contains code, prefabs, etc. for player-thrown item projectiles
    - **Props:** Contains static and non-interactable environmental decorations. If there is a prop that needs both a functional and non-functional version of the prefab (such as searchable objects), the meshes/materials/textures for the prop should be stored here and not in the FunctionalProps folder

- **Structures:** Contains prefab versions of buildings made with structure tile kits. May also contain baked mesh versions of these structures
- **StructureTiles:** Contains assets for modular building kits used to form the environment
- **SurfaceProperties:** Contains the audio, particles, and configuration assets for "surface properties", which defines how a collider reacts when hit or walked on
- **Surfaces:** Contains textures and materials used for structure tiles. Most of these are triplanar, but a few are object space. Snowy variants of surfaces are stored under a Snowy folder in each material category
- **Terrain:** Contains terrain textures/materials/shaders (but not terrain files themselves. Those are stored alongside the associated levels)
- **UI:** Contains everything related to UI, as well as scenes that are dedicated to UI interfaces (IE: TitleScreen, GameOver, etc.)

C# Coding Standards

- **Brackets**
  - Opening brackets go on new lines
  - When an if statement or loop only contains one line, don't use brackets at all. Just indent the line without brackets (not a super important rule)
- **Methods:** "MethodName" (pascal case)
  - **Awake:** Should be treated like a class constructor for components (EX: cached GetComponent calls would occur here)
  - **Start:** Should be treated as a secondary initialization call when the order of initialization can't be guaranteed on the first frame. If a class needs to access a singleton on initialization, it should occur HERE and NOT on Awake!
- **Fields**
  - **Private:** _"variableName" (camel case)
  - **Protected:** _"variableName" (camel case)
  - **Public:** "variableName" (camel case)
  - **Constant:** "VariableName" (pascal case)
  - **Auto Properties:** "VariableName" (pascal case)
    - Auto properties should be used in situations where it is better for clarity and length, as opposed to creating dedicated getter/setter methods
  - Do not include variable type in name (not a super important rule)
  - If a default value feels appropriate, this should be set directly alongside the field declaration, rather than in a constructor (otherwise the default value won't appear in the editor)
- **Exposed Editor Fields**
  - Don't always use public variables for exposing component fields in the editor
  - Private variables with a **[SerializeField]** attribute should be used in the following situations:
    - If it's used to assign a reference to a component or game object that is purely for aesthetic use and will never need accessed by other classes

        - If the field is only meant to be modified in the editor and initialized to a value, but never modified at runtime by another class
        - If it is being exposed for testing purposes but should otherwise be hidden from other scripts
- **Editor Attributes**
    - **Classes**
        - **[AddComponentMenu]** should always be used to customize the location of components in the add component dropdown menu
        - **[DisallowMultipleComponent]** if multiple instances of the component on an object would break something
        - **[RequireComponent(typeof("MonoBehaviourName"))]** if the script uses GetComponent on the object and assumes the presence of a particular component
            - Note that some components may have a large number of dependencies that would make this attribute excessively long. If this is the case, it can be omitted
    - **Fields**
        - **[Tooltip("TooltipContents")]** should be added if a component has a documentation comment associated with it. The contents of the tooltip can just be the same as the comment
- **GetComponent<"TypeName">():** This should ALWAYS be cached where possible. Never call this multiple times on Update
- **Comments:**
    - Comments should be used anywhere with potential ambiguity or confusion. They may also be used to visually distinguish long sections of code
    - **Documentation Comments (/// <summary> comments):** Should be written like proper sentences
        - Documentation comments should be used on all classes, fields, or methods that might be ambiguous as to its purpose or usage
        - It is optional to explicitly label the parameters or return value